Space Manufacturing: An Industry Deep Dive

Midterm Presentation

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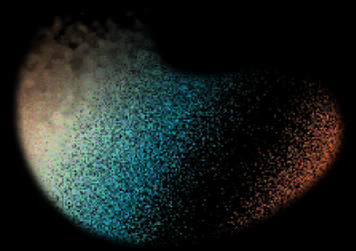
industrial production of high-value goods in Low-Earth Orbit



Key Products :

- Lossless Fiber Optic Cables
- Novel Drug Formulations
- High Performance Semiconductors

Macro Trends

Within 10 years time, there will be significant push towards an economically scalable space manufacturing industry

40x

in launch cost reductions,

$1.8T

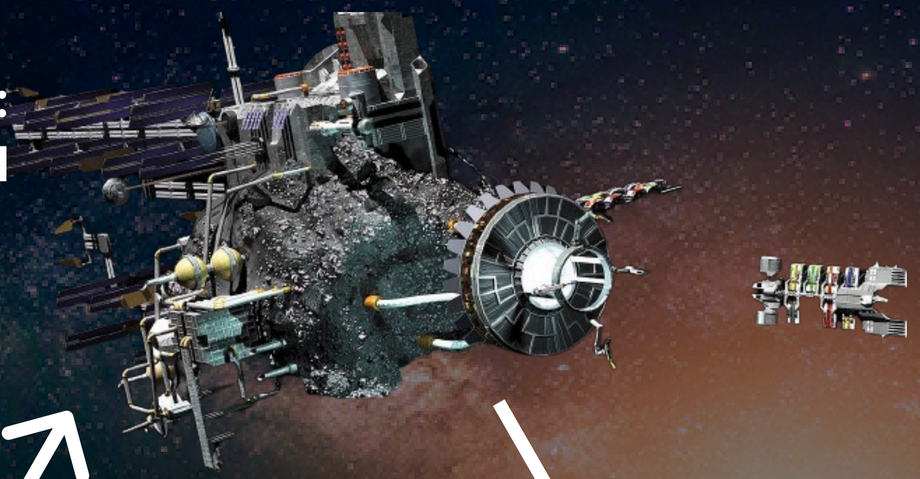
expected global economy in 10 years

90%

of value in industry is IP

Bottleneck

in-orbit manufacturing: 3D printing and furnaces that produce in microgravity



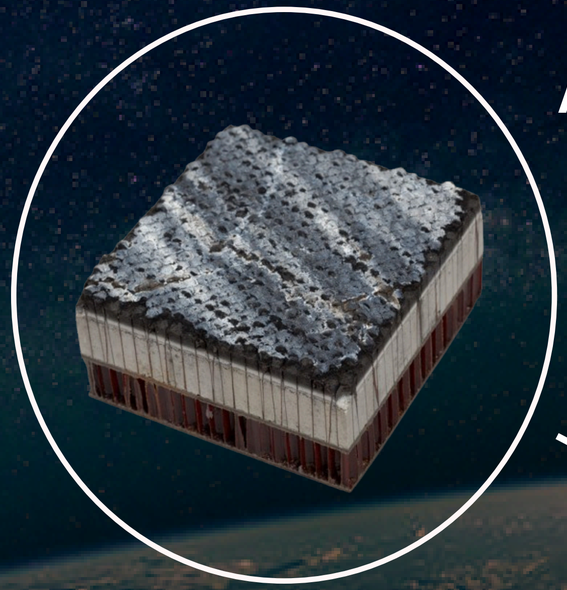
up-mass: getting things up in LEO

down-mass: how do we get the products back down?

Investment Direction

~$5B in VC investment

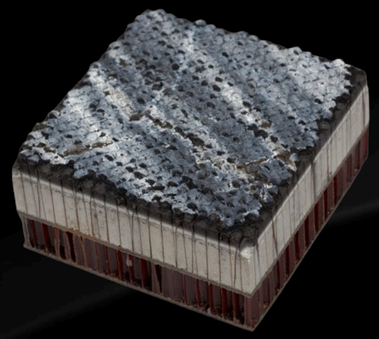
Critical Gap



Ablative Heat Shield

Heat Shields

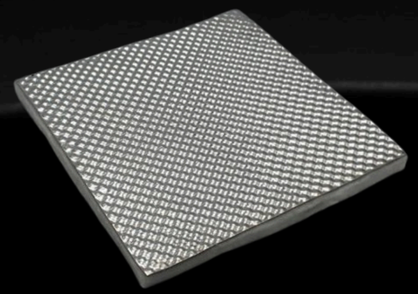
$2k/m²



Ablative

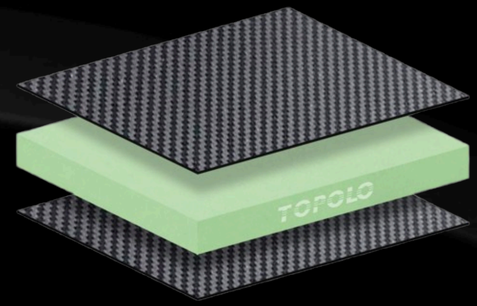
1950s

$10k/ft²



Silica Glass Fiber

$100k/ft²



Reinforced Carbon-Carbon

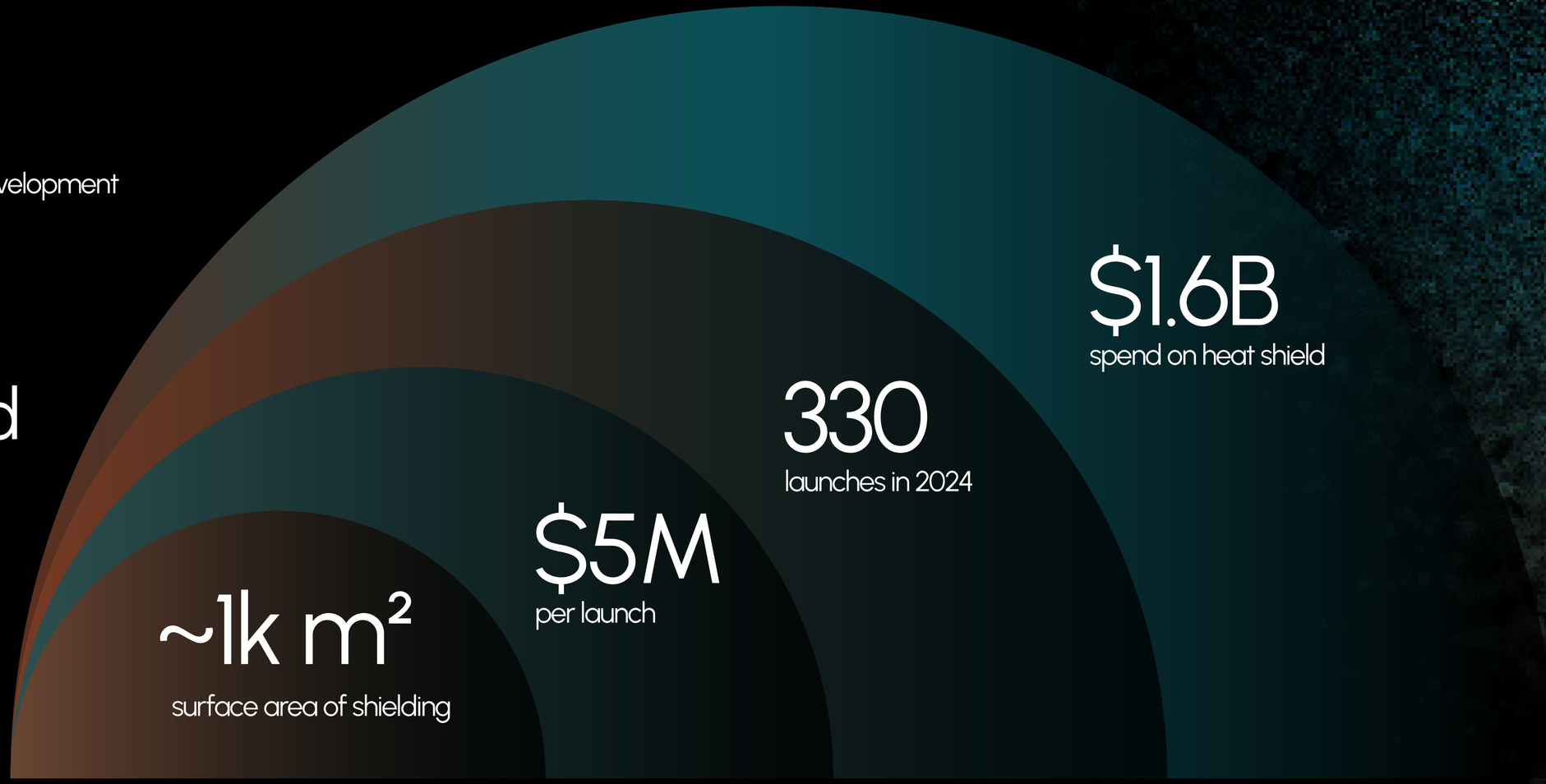
1980s

Problem

$1B+

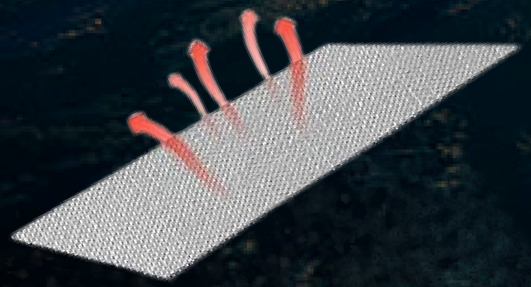
structural gap in heat shield development

heat shields are assumed to be a fixed cost

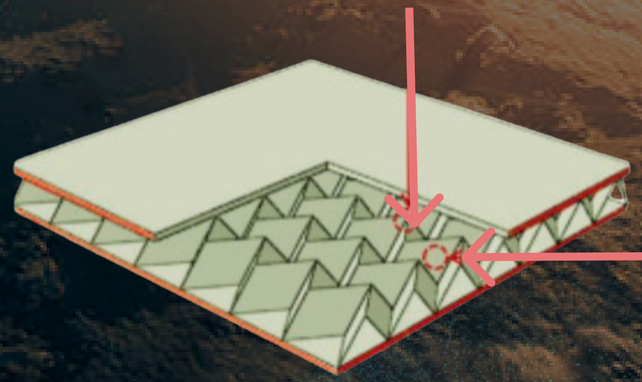


Breakthrough

reflects 95% of heat



membrane



rigid facet

Re-Usable Heat Shields

1. Unfolds after Launch-Stowage
2. Lightweight Fiber Weave
3. Drag Surface Deceleration
4. Radiative Heat Dissipation

One-Time

Low Mass

Turn Around

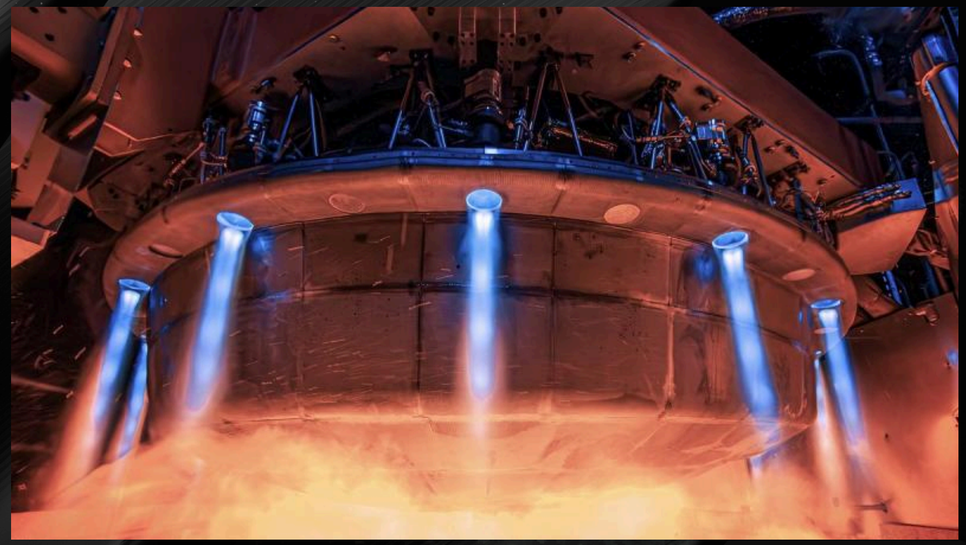
Thesis

Invest in Re-Usable Heat Shields

Space Forge

origami folding shuttlecock heat shields

Stoke Space



liquid hydrogen regenerative cooling

Thank You